Logic

Boolean algebra: Axioms of $\equiv$ for $\wedge$, $\vee$, $\neg$

Commutativity:

$$\phi \vee \psi \equiv \psi \vee \phi$$
$$\phi \wedge \psi \equiv \psi \wedge \phi$$

Idempotence:

$$\phi \vee \phi \equiv \phi$$
$$\phi \wedge \phi \equiv \phi$$

Associativity:

$$\phi \vee (\psi \vee \chi) \equiv (\phi \vee \psi) \vee \chi$$
$$\phi \wedge (\psi \wedge \chi) \equiv (\phi \wedge \psi) \wedge \chi$$

Complement:

$$\phi \vee \neg\phi \equiv \top$$
$$\phi \wedge \neg\phi \equiv \bot$$

Distributivity:

$$\phi \vee (\psi \wedge \chi) \equiv (\phi \vee \psi) \wedge (\phi \vee \chi)$$
$$\phi \wedge (\psi \vee \chi) \equiv (\phi \wedge \psi) \vee (\phi \wedge \chi)$$

De Morgan:

$$\neg(\phi \vee \psi) \equiv \neg\phi \wedge \neg\psi$$
$$\neg(\phi \wedge \psi) \equiv \neg\phi \vee \neg\psi$$

Identities:

$$\phi \vee \bot \equiv \phi$$
$$\phi \wedge \top \equiv \phi$$

Domination:

$$\phi \vee \top \equiv \top$$
$$\phi \wedge \bot \equiv \bot$$

Involution:

$$\neg\neg\phi \equiv \phi$$

Equivalence relation

A binary relation R is an *equivalence relation* if, for all elements x, y, z in its domain, R satisfies the following three properties:

- Reflexive: $x \; R \; x$
- Symmetric: $x \; R \; y$ implies $y \; R \; x$
- Transitive: $x \; R \; y$ and $y \; R \; z$ implies $x \; R \; z$

Equivalence relation R gives rise to *equivalence classes*.

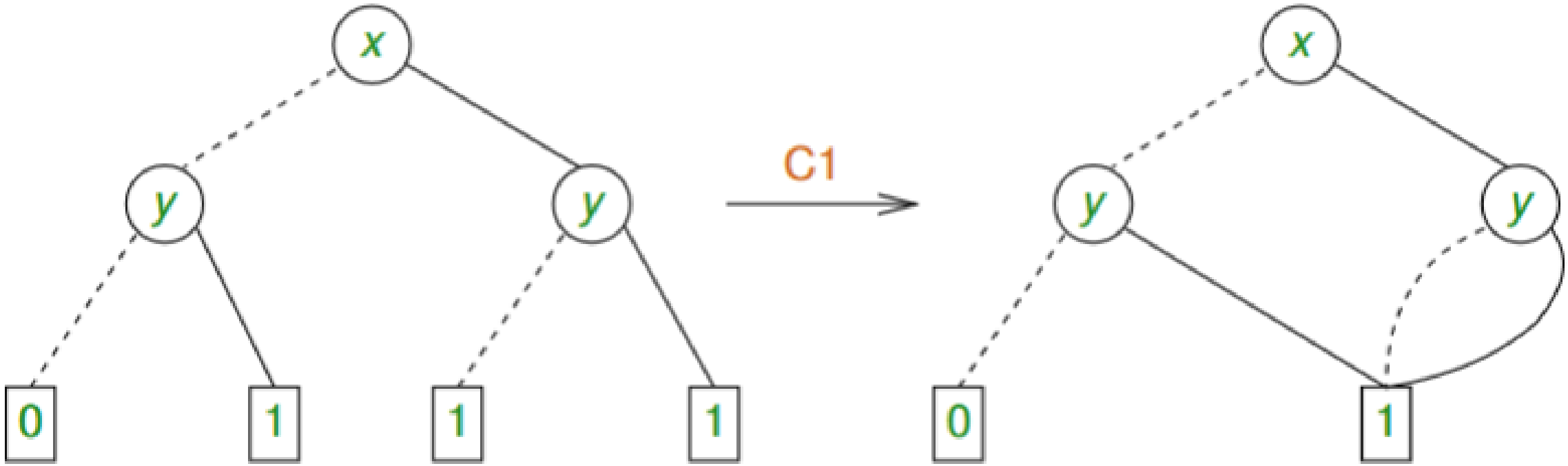
Within an equivalence class, all elements are related by R.

Elements in *different* equivalence classes are *not* related by R.

Reducing OBDDs - Rule C1

Rule C1: **Collapse identical leaves.**

Example: Consider the binary decision tree for $x \vee y$.



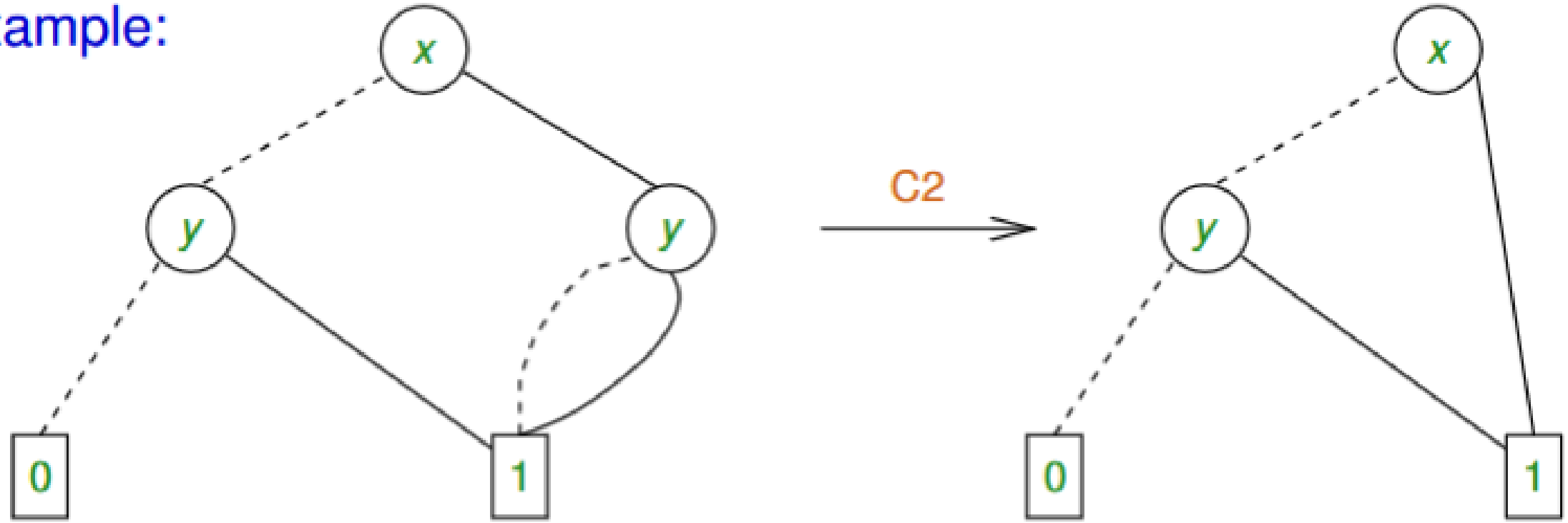
Reducing OBDDs - Rule C2

Rule C2: **Eliminate superfluous non-leaves.**

If both outgoing edges of a node n point to the same node n', then:

- eliminate n, and
- redirect its incoming edges to n'.

Example:

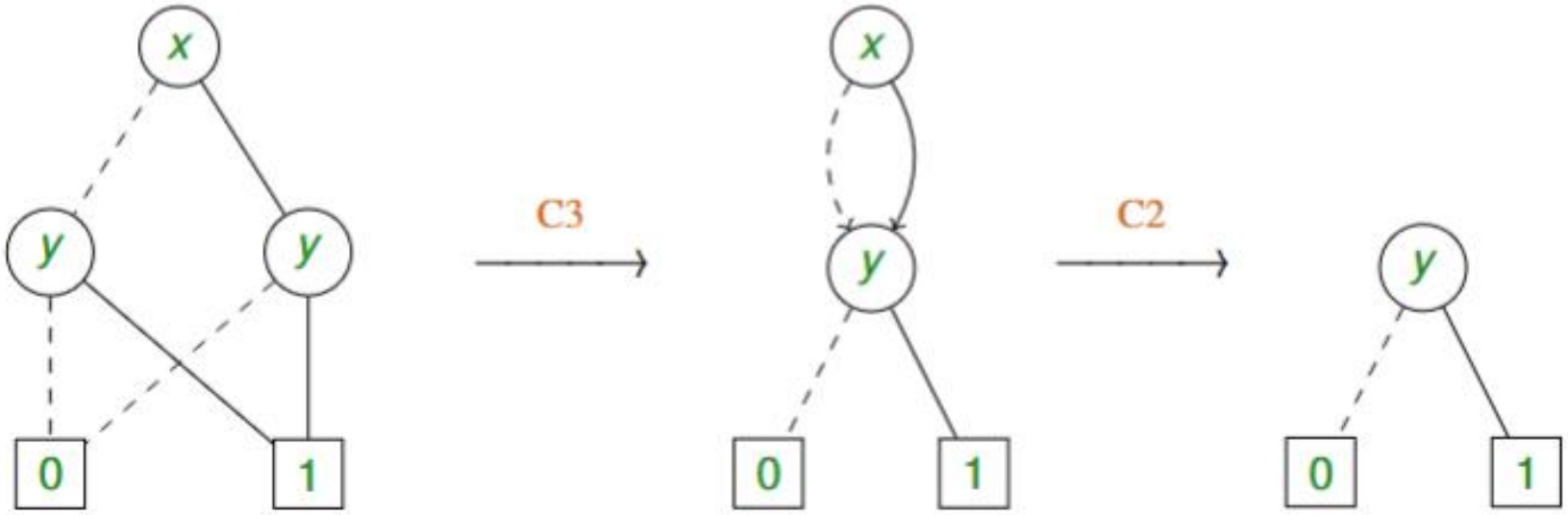


Reducing OBDDs - Rule C3

Rule C3: **Remove duplicate non-leaves.**

Collapse non-leaves associated with the same variable if their 0-edges and their 1-edges point to the same nodes.

Example: Consider the binary decision tree for $(x \wedge y) \vee (\neg x \wedge y)$, after collapsing its leaves with C1.



Some semantic equivalences

Important semantic equivalences in predicate logic:

$$\neg\exists x\,\phi \quad\equiv\quad \forall x\,\neg\phi$$

$$\neg\forall x\,\phi \quad\equiv\quad \exists x\,\neg\phi$$

$$\exists x\,(\phi \vee \psi) \quad\equiv\quad \exists x\,\phi \;\vee\; \exists x\,\psi$$

$$\forall x\,(\phi \wedge \psi) \quad\equiv\quad \forall x\,\phi \;\wedge\; \forall x\,\psi$$

$$\exists x\,\exists y\,\phi \quad\equiv\quad \exists y\,\exists x\,\phi$$

$$\forall x\,\forall y\,\phi \quad\equiv\quad \forall y\,\forall x\,\phi$$

Some non-equivalences

The following are not semantic equivalences:

$$\exists x\,(C(x) \wedge D(x)) \quad \not\equiv \quad \exists x\, C(x) \;\wedge\; \exists x\, D(x)$$

$$\forall x\,(C(x) \vee D(x)) \quad \not\equiv \quad \forall x\, C(x) \;\vee\; \forall x\, D(x)$$

$$\forall x\, \exists y\, K(x,y) \quad \not\equiv \quad \exists y\, \forall x\, K(x,y)$$

Sets

Injective and Surjective

- **Injective**: Focuses on uniqueness—no two inputs share the same output.
- **Surjective**: Focuses on coverage—every element in the codomain is an output.

Visual Analogy:

- Injective: Assigning unique ID numbers to people (no duplicates).
- Surjective: Ensuring every seat in a theater is occupied.